Math 110C Lesson 2 & Concepts Check (Quiz): Vectors (section 12.2)

- Due Sep 6 at 11:59pm
- Points 14
- Questions 14
- Available until Sep 7 at 11:59pm
- Time Limit None

Instructions

Math 110C	Vectors (section 12.2)

This corresponds to section 12.2 in your Calculus textbook.

- Here are the **lecture notes for you to fill out during our lesson (https://ccsf.instructure.com/courses/68102/files/14363598?wrap=1) (https://ccsf.instructure.com/courses/68102/files/14363598/download?download_frd=1)** .
- Here is a video for the complete lesson.

Math 110C 12.2 Vectors -- a complete lesson

- Read section 12.2 of your Calculus textbook.
- Check your understanding of the lesson by **taking the lesson concept check (Quiz)** below.

Now do **Math 110B HW for 12.2A & 12.2B** in the Assignment section of this module.

Attempt History

	Attempt	Time	Score
LATEST	Attempt 1	2,342 minutes	12 out of 14

Score for this quiz: 12 out of 14

Submitted Sep 6 at 2:55pm

This attempt took 2,342 minutes.

Correct answer

Question 1

1 / 1 pts

We denote a vector by printing a letter in boldface or by putting an arrow above the letter.

- (●) True
- ( ) False

Correct answer

Question 2

1 / 1 pts

A vector has both magnitude and direction.

- (●) True
- ( ) False

Correct answer

Question 3

1 / 1 pts

Two vectors are equivalent (or equal) if they have the same length and the same direction.

- (●) True
- ( ) False

Correct answer

Question 4

1 / 1 pts

The zero vector has a specific direction.

- ( ) True
- (●) False

Correct answer

Question 5

1 / 1 pts

To construct the sum of two vectors geometrically, we can use either the Triangle Law or the Parallelogram Law.

- (●) True
- ( ) False

Correct answer

Question 6

1 / 1 pts

Real numbers are scalars. They have no directions.

- (●) True
- ( ) False

Wrong answer

Question 7

0 / 1 pts

$(3,2)(3,2)$ and $<3,2><3,2>$ are equivalent representations of a vector.

- True
- False

Correct answer

Question 8

1 / 1 pts

$<3,2><3,2>$ is the position vector of the point $(3,2)(3,2)$.

- True
- False

Correct answer

Question 9

1 / 1 pts

The vector represented by the directed line segment with initial point $(2,-3,4)(2,-3,4)$ and terminal point $(-2,1,1)(-2,1,1)$ is $<0,-2,5><0,-2,5>$.

- True
- False

Correct answer

Question 10

1 / 1 pts

The magnitude or length of the vector **a** $=<a_1,a_2>=<a_1,a_2>$ is $\sqrt{(a_1)^2-(a_2)^2}\sqrt{\left(a_1\right)^2-\left(a_2\right)^2}$.

- True
- False

Correct answer

Question 11

1 / 1 pts

V_2V_2 is the set of all two-dimensional vectors.

- True

- False

Correct answer

Question 12

1 / 1 pts

The vectors **i, j,** and **k** are called the standard basis vectors in V_3 $\boldsymbol{V_3}$**.**

- True
- False

Correct answer

Question 13

1 / 1 pts

If a vector **a** $\neq \neq$ **0**, then we can find the unit vector that has the same direction as **a** by multiplying the vector **a** by its length,

- True
- False

Wrong answer

Question 14

0 / 1 pts

In the last example in our lesson, the tension vector $T_1 = T_1 \sin 50^\circ i + T_1 \cos 50^\circ j$

$T_1 = T_1 \sin 50^\circ i + T_1 \cos 50^\circ j$.

- True
- False

Quiz Score: 12 out of 14